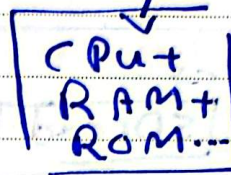
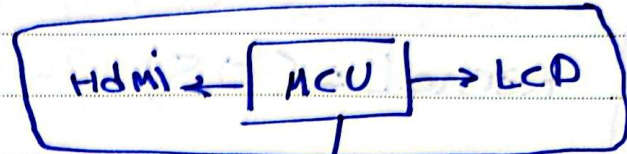


Embedded Systems (bare metal) vs Embedded Linux

Embedded system

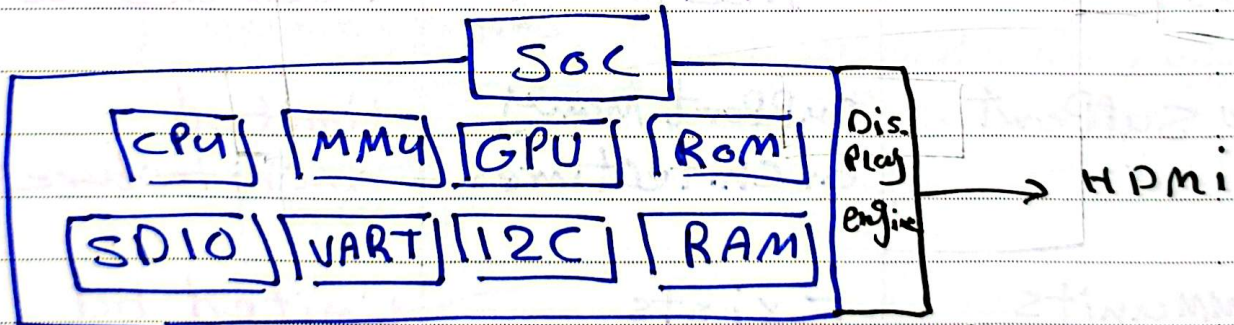


- direct interaction
- less execution time
- simple application
- limited for one purpose
- = dedicated for single functionality

* اختيار هو سيم سيم مدمج الهدف والهدف

SOC:- system on chip

* ده اختيار هو سيم سيم كامل افتر استرجه في كذا هدف



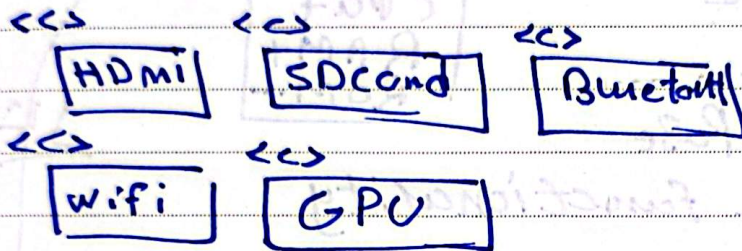
افتر ايفها في Board مع peripherals
 ده اختيار في آني بدل مائل واحد كابر مواصفات سيم يعني ال SOC
 عليها حاجة واستخدم الى غاية "وده اسرع واكبر"

لو انا هتعالج direct مع ال HW ده الى الى هتعالج ؟

- APP (GUI) has dependency hardware
- To many drivers

Complex application

Kernel = (OS)



ال OS هنا هو وسيط
يعزل عزل التطبيقات
وده هي حيل المستأجر
- مش محتاج السب drivers
من البداية هي موفرة
فال (OS)

- ان app مش مخد
على الهاردوير

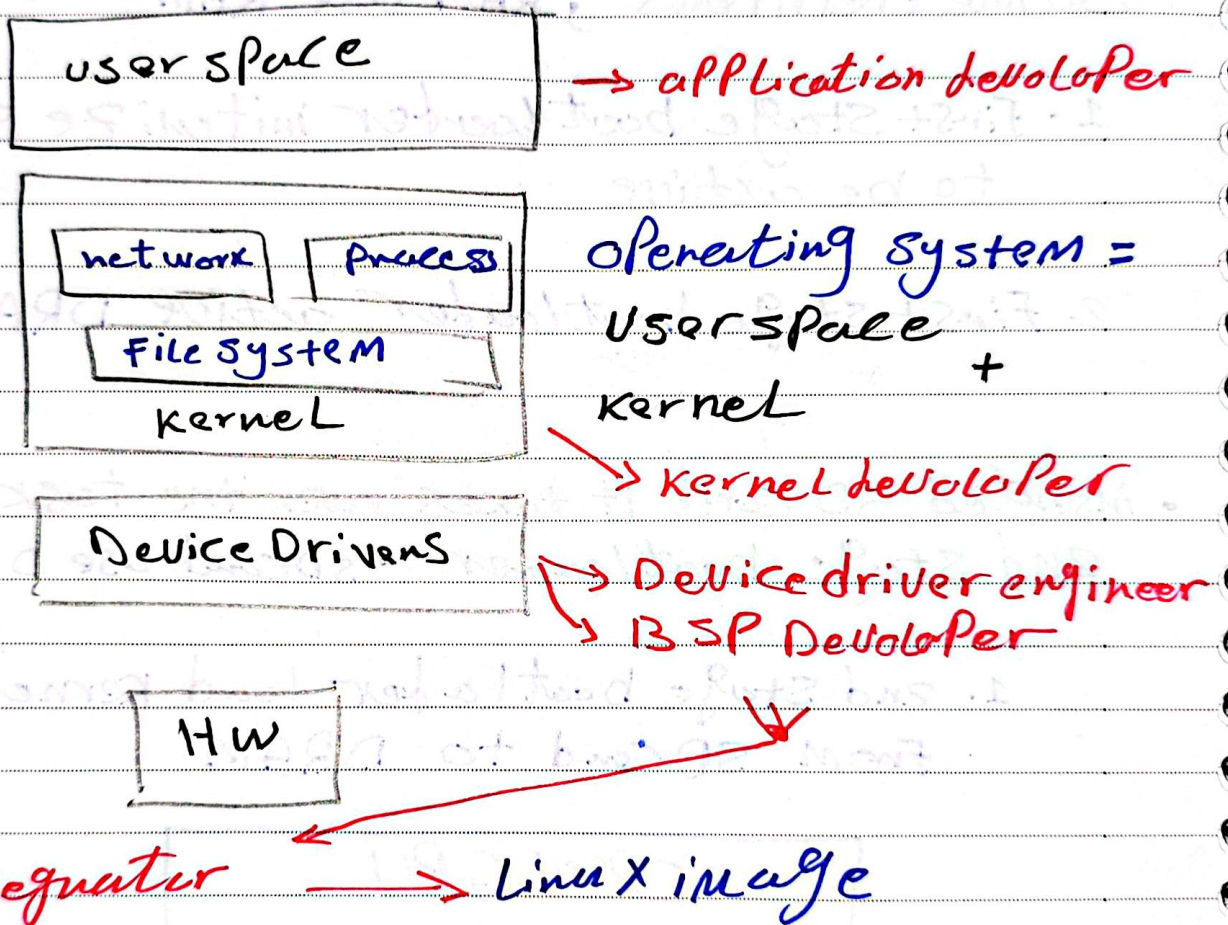
Point of Comparison	Embedded Linux	Embedded Windows
Cost	Free	Need Licence
Hw Support	Support Many architectures	Limited architectures
Communitis support	Exists	Limited by Microsoft
Foot Print	< 20	> 100 MB
	✓	✗

→ Embedded Linux:-

- complex application not a real-time
- Time to Market is faster
- Reduce cost - open source and free
- Manage resources - Driven

→ Linux Architecture:-

"ion titles"



Booting Process

SOC(CPU, GPU, RAM, ROM, SDIO, I2C)

- Power on starts → CPU, GPU, RAM, ROM
- Soc ROM contains or stores first stage Bootloader, Run on CPU to initialize some peripherals, "RAM use RAM"
 1. first stage boot loader initialize SDIO to be active
 2. first stage boot loader active DRAM
- inserted SD card it takes over the task of 2nd stage boot loader "SD card use DRAM"
 1. 2nd stage boot loader load kernel from SD card to DRAM

